Steel Dynamics, Inc.
Long Products Group
Structural and Rail Division
2601 S. County Road 700 East
Columbia City, Indiana 46725
(260) 625-8100 (260) 625-8950 FAX

Quality Steel 100% EAF Melted and Manufactured in the USA
ISO 9001:2015 and ABS Certified
CMTR complies with EN 10204 3.1.

CERTIFIED MILL TEST REPORT

Printed: **06 / 14 / 2021**
Produced: **06 / 08 / 2021**

Ship to:

Bill to:

GENERAL INFORMATION

Product	Wide Flange Beam
Size	W24X94 W610X140
Heat Number	B195531
Condition(s)	As-Rolled Fine Grained Fully Killed No Weld Repair

SPECIFICATIONS

Standards	Grades*
ASTM A6/A6M - 19	
» ASTM A992/A992M - 20	A992 / A992M
ASTM A572/A572M - 21	A572 gr50/gr345
ASTM A709/A709M - 18	A709 gr50/gr345
AASHTO M270M/M270 - 20	M270 gr345/gr50
CSA G40.21-13	50WM/345WM
ASTM A36/A36M - 19	A36 / A36M

*SDI-MULTI meets the requirements of ASTM A992, A572-50, A709-50, A529-50, A36 and A709-36; AASHTO M270-50 and M270-36; CSA 300W, 345W and 350W.

SHIPMENT DETAILS

BOL # 0000658194 - 47000.00 lbs

Bundle / ASN #	Length	pcs	Cust PO | Job/Reference	
022995165	50' 0"	2	21004-03	ECB 3.0
022995167	50' 0"	2	21004-03	ECB 3.0
022995168	50' 0"	2	21004-03	ECB 3.0
022995171	50' 0"	2	21004-03	ECB 3.0
022995163	50' 0"	2	21004-03	ECB 3.0

CHEMICAL ANALYSIS (weight percent)

C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	V	Nb/Cb	Al	N	B	*CE1	*CE2	*CE3	*PCM	*I	Analysis Type
.08	1.37	.013	.018	.22	.35	.14	.19	.04	.012	.047	.002	.002	.0136	.0005	.40	.433	.34	.19	5.95	Heat

MECHANICAL TESTING

Test	Yield (fy) Strength ksi / MPa	Tensile (fu) Strength ksi / MPa	fy / fu ratio	% Elong. {8" gage}
1	60 / 415	80 / 550	.76	26
2	62 / 425	80 / 550	.77	22
3				
4				

CHARPY IMPACT TESTS (available only when specified at time of order)

Test	Temp F / C	Absorbed Energy ft-lbf / J Specimen 1	Specimen 2	Specimen 3	Average	Minimum
1						
2						
3						
4						
5						
6						
7						

Notes: *Calculated Chemistry Values: Carbon Equivalents (C1, C2, C3, PC), Corrosion Index (I) {ASTM G101}= 26.01(Cu)+3.88(Ni)+1.20(Cr)+1.49(Si)+17.29(P)-7.29(Cu)(Ni)-9.10(Ni)(P)-33.39(Cu²) **Pcm**{AWS} = C+Si/30+Mn/20+Cu/20+Ni/60+Cr/20+mo/15+V/10+5B
CE1 {IIW}=C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15 **CE2** {AWS}=C+(Mn+Si)/6+(Cr+Mo+V)/5+(Ni+Cu)/15 **CE3** {CET} = C + (Mn/6) + (Si/24) + (Cr/5) + (Ni/40) +(Mo/4) + (V/14)

I hereby certify that the material described herein has been made to the applicable specification by the electric arc furnace/continuous cast process and tested in accordance with the requirements of American Bureau of Shipping Rules with satisfactory results.

Signed:

ABS CERTIFICATION

I hereby certify that the content of this report are accurate and correct. All tests and operations performed by this material manufacturer are in compliance with the requirements of the material specifications and applicable purchaser designated requirements.

Signed: **Valoree Varick** Valoree Varick
Quality Manager

State of Indiana, County of Whitley Sworn to and subscribed before me

this __________ day of ____________________

Signed: ____________________ My commission expires: __________
Notary Public

Form F-6100-002-054 rev 8

ASTM A6 - 14.6: A signature is not required on the test report; however, the document shall clearly identify the organization submitting the report. Notwithstanding the absence of a signature, the organization submitting the report is responsible for the content of the report

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GENERAL INFORMATION

Product	Wide Flange Beam
Size	W24X94 W610X140
Heat Number	A195536
Condition(s)	As-Rolled Fine Grained Fully Killed No Weld Repair

SPECIFICATIONS

Standards	Grades*
ASTM A6/A6M - 19	
» ASTM A992/A992M - 20	A992 / A992M
ASTM A572/A572M - 21	A572 gr50/gr345
ASTM A709/A709M - 18	A709 gr50/gr345
AASHTO M270M/M270 - 20	M270 gr345/gr50
CSA G40.21-13	50WM/345WM
ASTM A36/A36M - 19	A36 / A36M

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SHIPMENT DETAILS

BOL # 0000658194 - 47000.00 lbs

Bundle / ASN #	Length	pcs	Cust PO	Job/Reference
022995154	50' 0"	2	21004-03	ECB 3.0
022995157	50' 0"	2	21004-03	ECB 3.0
022995159	50' 0"	2	21004-03	ECB 3.0
022995160	50' 0"	2	21004-03	ECB 3.0
022995152	50' 0"	2	21004-03	ECB 3.0

CHEMICAL ANALYSIS (weight percent)

C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	V	Nb/Cb	Al	N	B	*CE1	*CE2	*CE3	*PCM	*I	Analysis Type
.09	1.36	.012	.021	.23	.32	.12	.16	.04	.011	.046	.001	.002	.0139	.0006	.40	.434	.34	.20	5.82	Heat

MECHANICAL TESTING

Test	Yield (fy) Strength ksi / MPa	Tensile (fu) Strength ksi / MPa	fy / fu ratio	% Elong. {8" gage}
1	62 / 425	80 / 550	.78	24
2	63 / 435	80 / 550	.78	25
3				
4				

CHARPY IMPACT TESTS (available only when specified at time of order)

Test	Temp F / C	Absorbed Energy ft-lbf / J Specimen 1	Specimen 2	Specimen 3	Average	Minimum
1						
2						
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5						
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7						

Notes: *Calculated Chemistry Values: Carbon Equivalents (C1, C2, C3, PC), Corrosion Index (**I**) {ASTM G101}= 26.01(Cu)+3.88(Ni)+1.20(Cr)+1.49(Si)+17.29(P)-7.29(Cu)(Ni)-9.10(Ni)(P)-33.39(Cu²) **Pcm**{AWS} = C+Si/30+Mn/20+Cu/20+Ni/60+Cr/20+mo/15+V/10+5B
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Signed: **Valoree Varick** Valoree Varick
Quality Manager

Form F-6100-002-054 rev 8

State of Indiana, County of Whitley Sworn to and subscribed before me

this __________ day of ______________________

Signed: ______________________ My commission expires:__________
Notary Public

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